

LECTURE No. 3

BUSINESS PROCESSES MODELING

List of Topic to be Discuss in this Lecture

- ✓ Business Functions
- ✓ Business Processes
- ✓ Relationship between Business Processes and Business Functions
- ✓ Business Processes Modeling Notations
- ✓ Case Studies

INTRODUCTION

Understanding **business processes** and **business functions** is essential for effectively managing an organization and **optimizing** its operations. Both terms are interconnected but focus on different aspects of how an organization operates.

BUSINESS FUNCTIONS

A **business function** refers to a group of related activities or tasks that are specialized and typically handled by a specific department within an organization.

Business functions focus on the specific areas of expertise or responsibility that contribute to the overall success of the organization.

Characteristics of Business Functions:

- **Departmental:** Business functions are typically confined to a specific department.
- **Specialized:** Each function has specialized knowledge and skills necessary for executing specific tasks.

Sir Syed University of Engineering and Technology
Department of Computer Science and Information Technology
BS (Computer Science / Information Technology)

- **Ongoing:** Business functions are continuous and support the ongoing operations of the business.

Major Business Functions:

1. Finance and Accounting:

- Manages financial resources, including budgeting, accounting, and financial reporting.

Example Tasks: Preparing financial statements, managing cash flow, conducting audits, and payroll processing.

2. Human Resources (HR):

- Manages employee-related activities such as hiring, training, payroll, benefits, and compliance with labor laws.

Example Tasks: Recruitment, performance reviews, employee relations, and benefits management.

3. Sales and Marketing:

- Focuses on promoting products or services and generating revenue.

Example Tasks: Developing marketing strategies, conducting market research, advertising, managing customer relationships, and closing sales.

4. Operations/Production:

- Oversees the production of goods and services, ensuring efficiency and quality.

Example Tasks: Managing supply chains, production scheduling, inventory control, and quality assurance.

5. IT and Technology:

- Supports the technological needs of the business, managing information systems, infrastructure, and digital solutions.

Example Tasks: Maintaining hardware and software, implementing new technologies, cyber security, and supporting business applications.

Sir Syed University of Engineering and Technology
Department of Computer Science and Information Technology
BS (Computer Science / Information Technology)

Examples of Business Functions:

1. Sales Function:

- A sales department is responsible for finding and acquiring customers, selling products or services, and maintaining customer relationships.

Example: A sales team contacting potential clients, delivering product presentations, and closing deals.

2. HR Function:

- Human resources manage the recruitment and retention of employees, ensuring the organization has the talent it needs.

Example: HR managers interviewing candidates, organizing employee training sessions, and conducting performance evaluations.

3. Finance Function:

- The finance department handles all financial activities, such as tracking expenses, managing budgets, and generating financial reports.

Example: A finance team preparing quarterly financial statements and ensuring tax compliance.

BUSINESS PROCESSES

A **business process** is a set of activities or tasks designed to achieve a specific organizational goal. These processes are systematic, repeatable, and designed to deliver value to customers, either internal or external. Business processes are often cross-functional, involving multiple departments within an organization.

Characteristics of Business Processes:

- 1. Structured:** Business processes are typically well-defined, with clear steps and workflows.
- 2. Repeatable:** Processes are designed to be executed repeatedly to ensure consistency.
- 3. Goal-Oriented:** The primary purpose of a business process is to accomplish a specific objective.

4. **Measurable:** Business processes are usually monitored and evaluated based on key performance indicators (KPIs).

Core, Support, and Management Business Processes

- **Core Processes:** Directly deliver the organization's product or service (e.g., Order-to-Cash).
- **Support Processes:** Assist core processes without directly contributing to the product (e.g., Procure-to-Pay for raw materials).
- **Management Processes:** Oversee operations to align with organizational goals (e.g., strategic planning).

Importance of Standardization in Business Processes

- **Consistency:** Standardized workflows ensure all employees follow the same guidelines, reducing ambiguity and improving task execution. For example, a uniform Procure-to-Pay process streamlines vendor interactions and approval steps.
- **Minimizing Errors:** Clear, repeatable processes reduce the chances of mistakes. In the Order-to-Cash process, a standardized order entry template ensures accurate data for smoother downstream tasks like shipping and billing.
- **Resource Optimization:** Standardization eliminates redundancies, allowing for better resource allocation. For example, a Recruit-to-Retire HR process prevents overlap in onboarding and training, saving time and effort.
- **Outcome:** Standardization leads to more efficient workflows, better coordination between departments, reduced costs, and higher organizational performance.

Sir Syed University of Engineering and Technology
Department of Computer Science and Information Technology
BS (Computer Science / Information Technology)

Examples of Business Processes:

1. Order-to-Cash Process (Core Process):

- This process encompasses all activities from receiving a customer order to receiving payment.

Steps include: Order entry, order fulfillment, shipping, invoicing, and payment collection.

Example: An e-commerce company receives a customer order, processes it, ships the product, generates an invoice, and collects payment.

2. Procure-to-Pay Process (Support Process):

- This involves the process of acquiring goods or services, from requesting a purchase to making a payment to the supplier.

Steps include: Purchase requisition, vendor selection, purchase order creation, goods receipt, and payment.

Example: A manufacturing company purchasing raw materials, from requisitioning materials to paying suppliers.

3. Recruit-to-Retire Process (Support Process):

- This process covers the entire employee lifecycle, from recruitment to retirement.

Steps include: Recruitment, onboarding, performance management, training, and retirement or exit management.

Example: A company's HR team hires an employee, manages their career progression, and facilitates their retirement.

Importance of Business Processes:

- **Efficiency:** Well-defined processes help streamline operations and reduce inefficiencies.
- **Quality Control:** Standardizing processes ensures consistency and high quality.
- **Customer Satisfaction:** Efficient processes lead to timely delivery and higher customer satisfaction.
- **Scalability:** Businesses can grow by replicating efficient processes across departments or regions.

RELATIONSHIP BETWEEN BUSINESS FUNCTIONS AND BUSINESS PROCESSES:

- **Business Functions Support Business Processes:**

Each business function plays a role in completing a business process. For example, the HR function manages the recruit-to-retire process, and the finance function supports the procure-to-pay process.

- **Cross-Functional Collaboration:**

Business processes often span multiple functions. For example, the order-to-cash process involves the sales, finance, and logistics functions working together to fulfill an order and collect payment.

Examples of Business Processes Involving Multiple Functions:

1. **Order Fulfillment Process (Cross-Functional):**

- **Departments Involved:** Sales, finance, warehouse, and logistics.
Steps: The sales team enters an order, the finance team processes payment, and the warehouse and logistics teams fulfill and ship the product.

2. Product Development Process (Cross-Functional):

- **Departments Involved:** Research and development (R&D), marketing, and operations.
Steps: The R&D department designs a new product, the marketing team researches customer needs, and the operations team handles the production process.

SUMMARY:

- **Business processes** are the workflows that organizations follow to achieve specific objectives, often involving multiple departments.
- **Business functions** are the specialized departments or areas within an organization responsible for specific tasks that support these processes.
- **Business processes** often involve the collaboration of multiple functions to deliver value to customers and meet organizational goals.
- Effective **business processes** and well-managed business functions lead to greater organizational efficiency, improved product and service quality, and increased customer satisfaction.

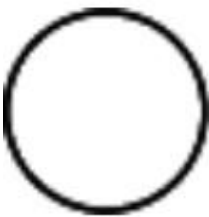
Business Process Modeling Notation (BPMN 2.0)

For every type of **process**, there is a **flowchart**. For **business**, there's even an entire process flowchart language. It's called **Business Process Model and Notation**, or simply, **BPMN 2.0**.

Common BPMN Symbols

BPMN event types

Events represent an event in a business process.



Start event symbol - Signals the first step of a process.



Intermediate event symbol - Represents any event that occurs between a start and end event.



End event symbol - Signals the final step in a process.

BPMN event Symbols

Each of these **BPMN event symbols** can be styled to represent the specific details of their process. The examples below are contained within start event symbols, but they can be combined with any event type. Common event symbols represent the following circumstances:



Message symbol - Triggers the process, facilitates intermediate processes, or finishes the process.



Timer symbol - A time, date, or recurring time and date triggers the process, aids intermediate processes, or completes the process.



Escalation symbol - A step reacts on an escalation and flows to another role in the organization. This event is only used within an event sub-process. An escalation occurs when someone at a higher level of responsibility within the organization becomes involved in a process.



Conditional symbol - A process begins or continues when a business condition or business rule is met.



Link symbol - A sub-process that is part of a larger process.



Error symbol - A caught error at the start, middle, or end of a process. An event sub-process with an error trigger will always interrupt its containing process.

Sir Syed University of Engineering and Technology
Department of Computer Science and Information Technology
BS (Computer Science / Information Technology)



Cancel symbol - Reacts on a transaction that was canceled within a sub-process. In an end event, the cancel symbol represents the triggered cancellation of a process.



Compensation symbol - A refund that's triggered when operations partially fail.



Signal symbol - A signal that communicates across different processes. A signal symbol can begin a process, facilitate it, or complete it.



Multiple symbol - Multiple triggers initiating a process.



Parallel multiple symbol - A process instance that doesn't start, continue, or end until all possible events have occurred.



Terminate symbol - Triggers the immediate termination of a process step. All related instances are terminated at the same time.

BPMN activity symbols

Activities describe the kind of work being done in a particular process instance. There are four **BPMN activity types**: tasks, sub-processes, transactions, and call activities.



Task symbol - The most basic level of an activity and cannot be broken down further. For example, a morning routine process might involve the task of turning on your computer.



Sub-process symbol - Is a group of tasks that fit together particularly well. There are two different views of the sub-process. One is the collapsed view, which has an expandable plus sign to show more details. The other view is an expanded sub-process view, which is large enough to house all the tasks that fully describe the sub-process.



Transaction symbol - Is a specialized sub-process that involves payment.



Call symbol - Is a global sub-process that is reused at various points in the business flow.

BPMN gateway symbols

In a BPMN diagram, **gateway symbols** that separate and recombine flows. There are several types of gateways:



Exclusive symbol - Evaluates the state of the business process and, based on the condition, breaks the flow into one or more mutually exclusive paths. For example, one report will be written if a supervisor grants approval; no report will be generated if the supervisor withholds approval.



Event-based symbol - An event-based gateway is similar to an exclusive gateway—both involve one path in the flow. In the case of an event-based gateway, however, you are evaluating which event has occurred, not which condition is being met. For example, you might wait to send an email until the CEO has arrived at the office. If the CEO doesn't arrive, the email will remain unsent.



Parallel symbol - Differs from other gateways because it's not dependent on conditions or events. Instead, parallel gateways are used to represent two concurrent tasks in a business flow. An example is a marketing department generating new sales leads and contacting existing leads at the same time.



Inclusive symbol - Breaks the process flow into one or more flows. For example, an inclusive gateway could involve business actions taken based on survey results. One process may be triggered if the consumer is satisfied with Product A. Another flow is triggered when the consumer indicates that they are satisfied with product B, and a third process is triggered if they aren't satisfied with A.



Exclusive event-based symbol - Starts a new process instance with each occurrence of a subsequent event.



Complex symbol - These gateways are only used for the most complex flows in a business process. An ideal use case for the complex gateway is when you need multiple gateways to describe the business flow.



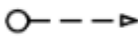
Parallel event-based symbol - As the name suggests, this gateway is similar to a parallel gateway. It allows for multiple processes to happen at the same time, but unlike the parallel gateway, processes are event-dependent.

Connecting objects in a BPMN diagram

Connecting objects are lines that connect BPMN flow objects. There are three different types: sequence flows, message flows, and associations.



Sequence flow symbol - Connects flow objects in proper sequential order.



Message flow symbol - Represents messages from one process participant to another.



Association symbol - Shows relationships between artifacts and flow objects.

Swimlanes in a BPMN diagram

Swimlanes are used to organize aspects of a process in a BPMN diagram. Swimlanes visually group objects into lanes, with each aspect of the process added to a separate lane. These elements can be arranged either horizontally or vertically. Swimlanes not only organize activities into separate categories, they can reveal delays, inefficiencies, and the workers responsible for each step in a process.



Artifacts in a BPMN diagram

Artifacts represent information relevant to the model but not to individual elements within the process. The three **artifact types** are annotations, groups, and data objects that can be used in a BPMN diagram. All three are used to augment and describe a BPMN process.

Annotations allow the modeler to describe additional flow parts of the model or notation.

Groups organize tasks or processes that have significance in the overall process.

Data objects represent data placed into the process, data resulting from the process, data that needs to be collected, or data that must be stored.



Data input symbol - Represents data requirements that tasks in the business process depend on.



Data output symbol - Demonstrates information produced as the result of a business process.

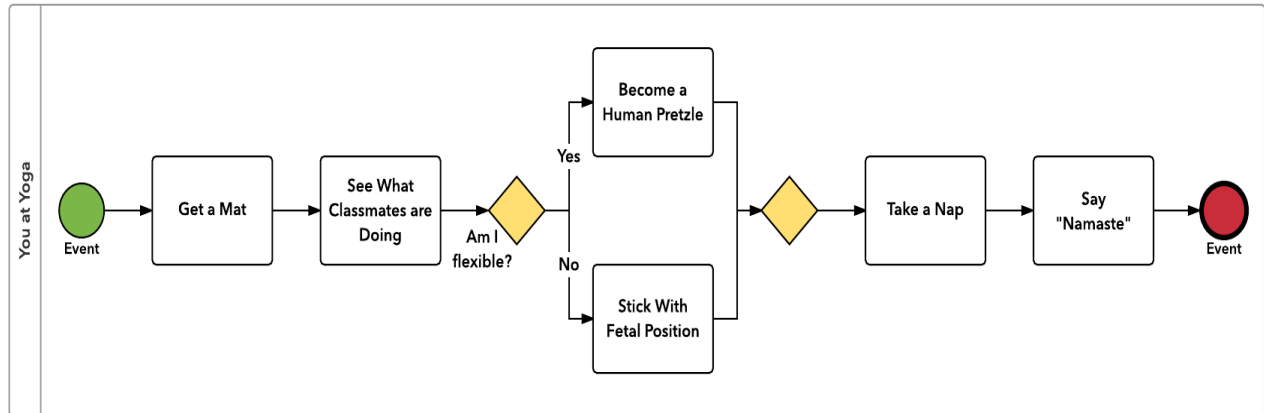


Data collection symbol - Signifies information collected within a business process.

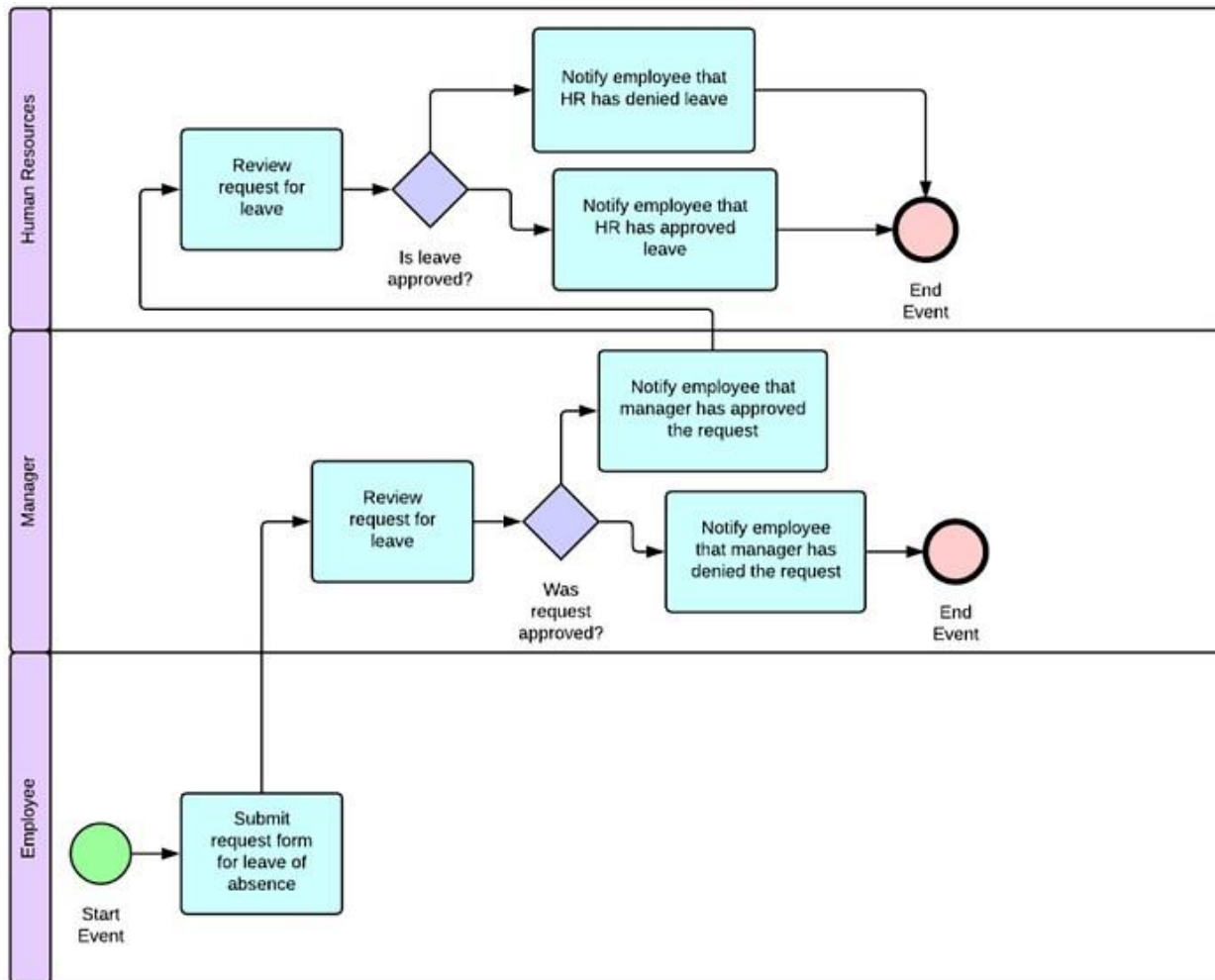


Data storage symbol - Represents the ability to store or access data that's associated with a business process.

Case Study 1: You're Yoga Process



Case Study 2: Leave Request to HR Process



Case Study 3: Order Management Process

